

Introduction to spatial data in R for paleontologists

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2026-08-30



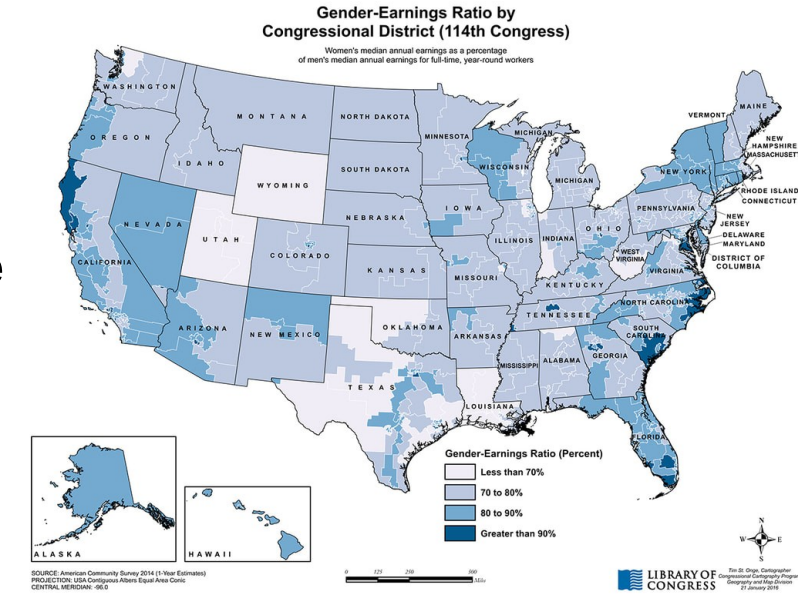
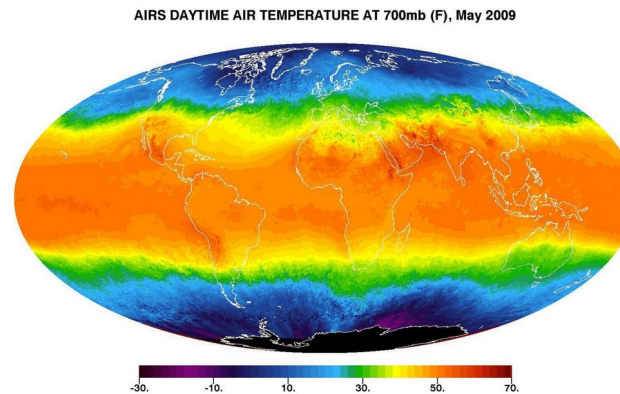
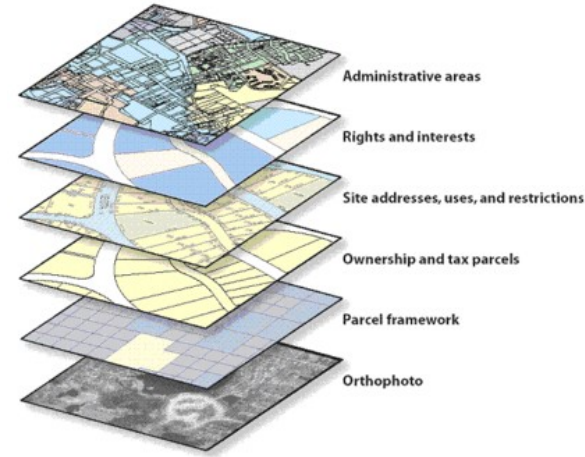
Basic Concepts

Where, what, why?

GIS basics

G(eographic) I(nformation) S(ystems/software) are everywhere

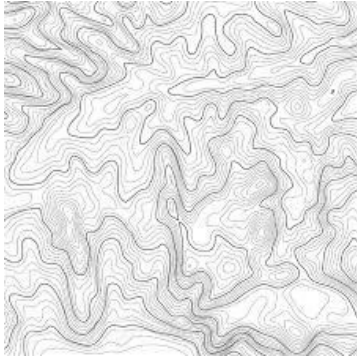
- Science
- Civil Engineering
- Aeronautics
- Military
- Economics
- etc.



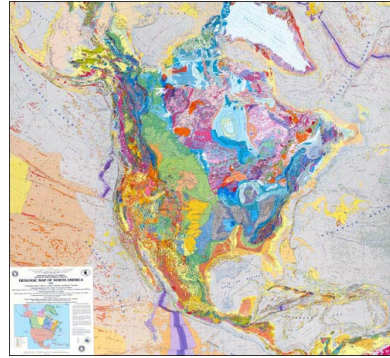
Vital in Earth Sciences

G(eographic) I(nformation) S(ystems) are everywhere

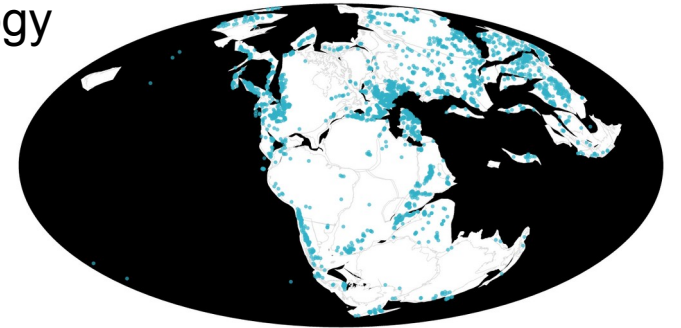
- topography



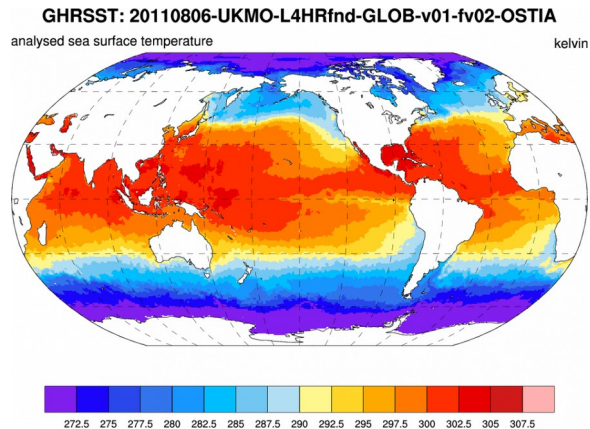
- geology



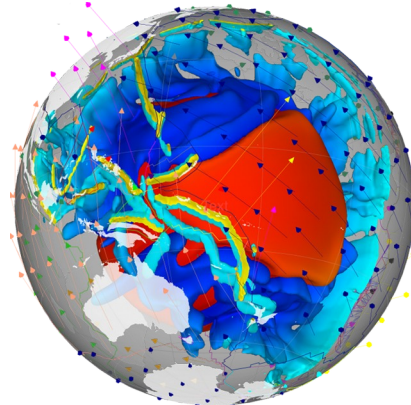
- paleontology



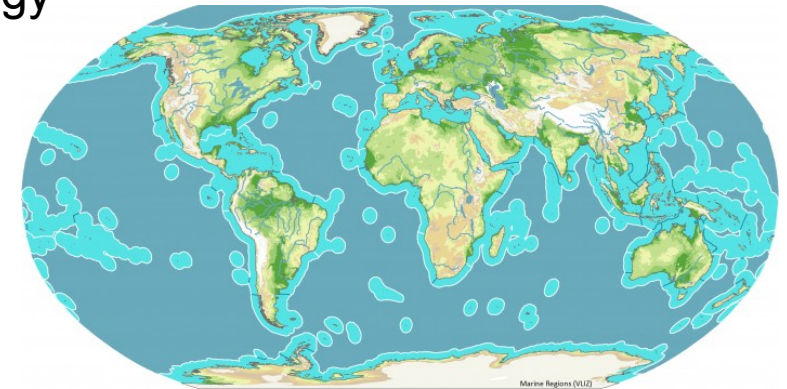
- climate



- geophysics



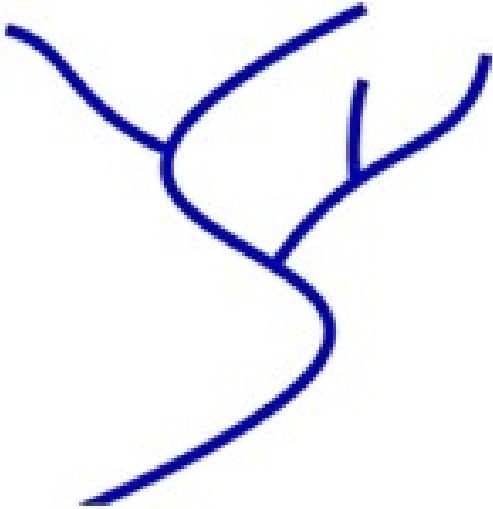
- ecology



Basic data types

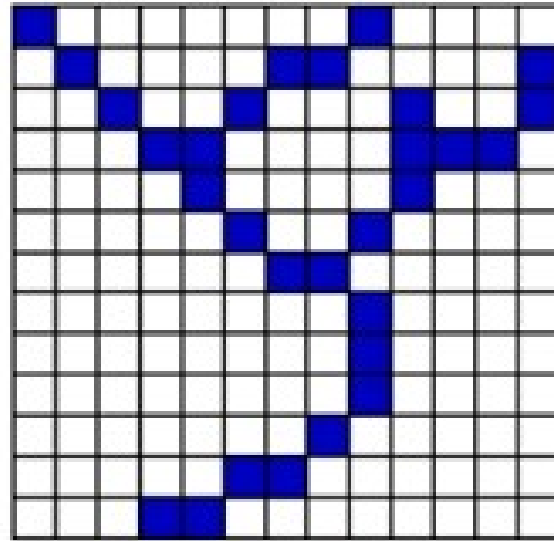
- Similar to computer graphics
- Sometimes in 3D

Object view



Vector

Field view

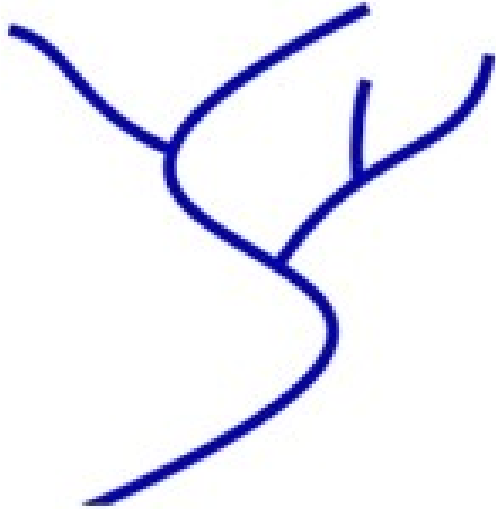


Raster

Basic data types

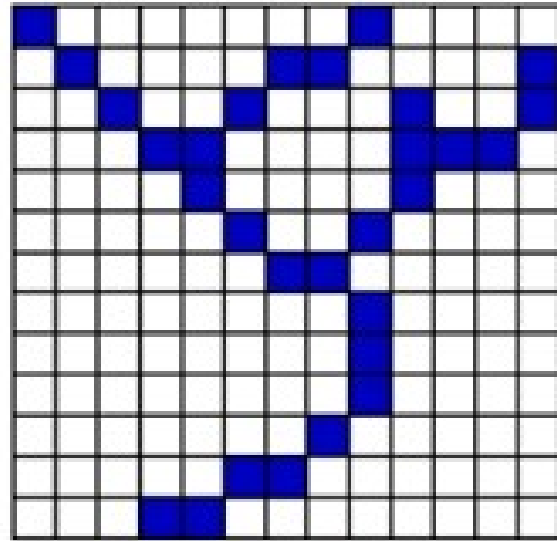
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Object view



Vector

Field view



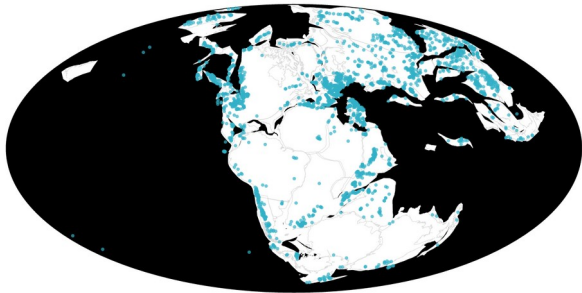
Raster

There is one twist:



Vector data

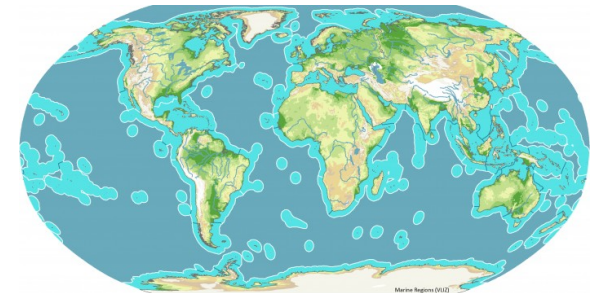
- Structure is defined with points that have coordinates
- Spherical coordinates, Cartesian too, or projection-specific
- Additional attributes are rendered to the entities
- 3 primary kind



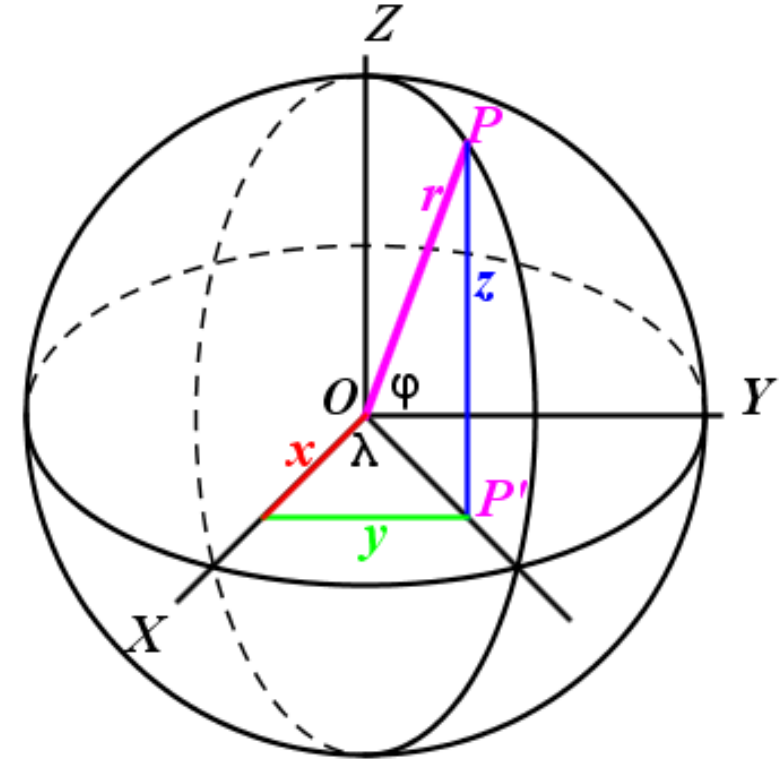
Points



Lines

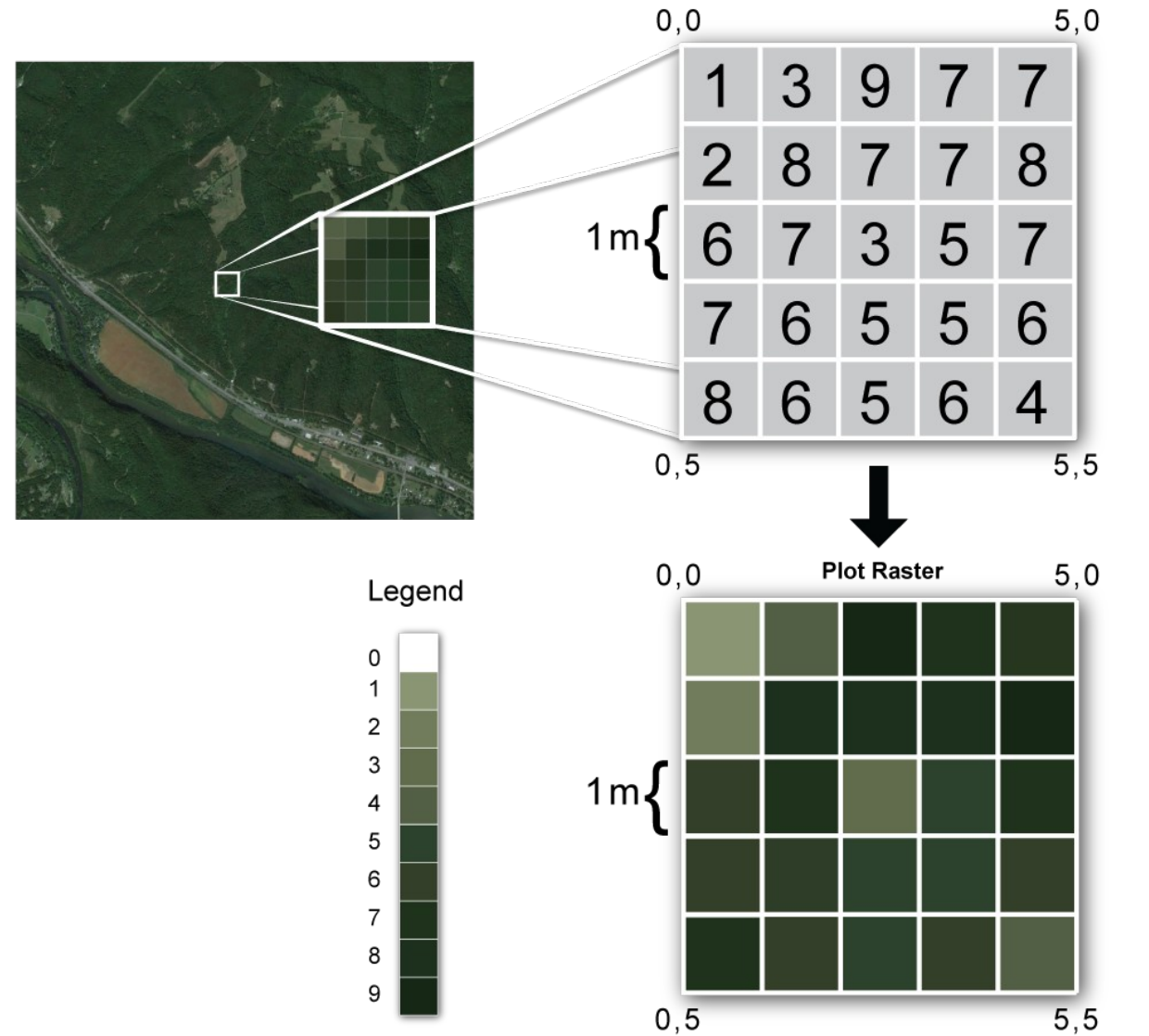


Polygons (shapes)



Raster data

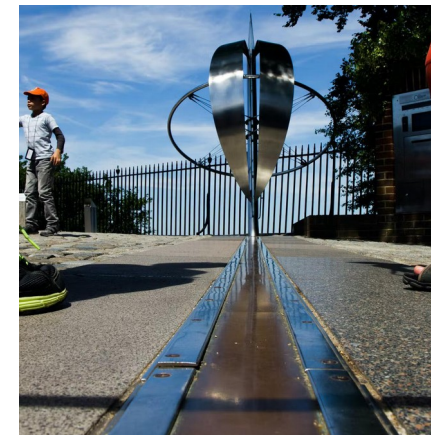
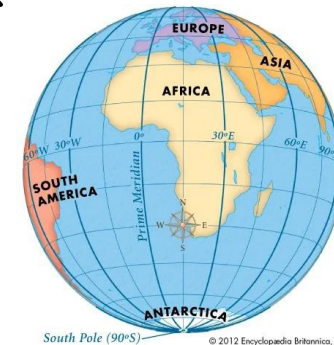
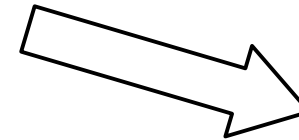
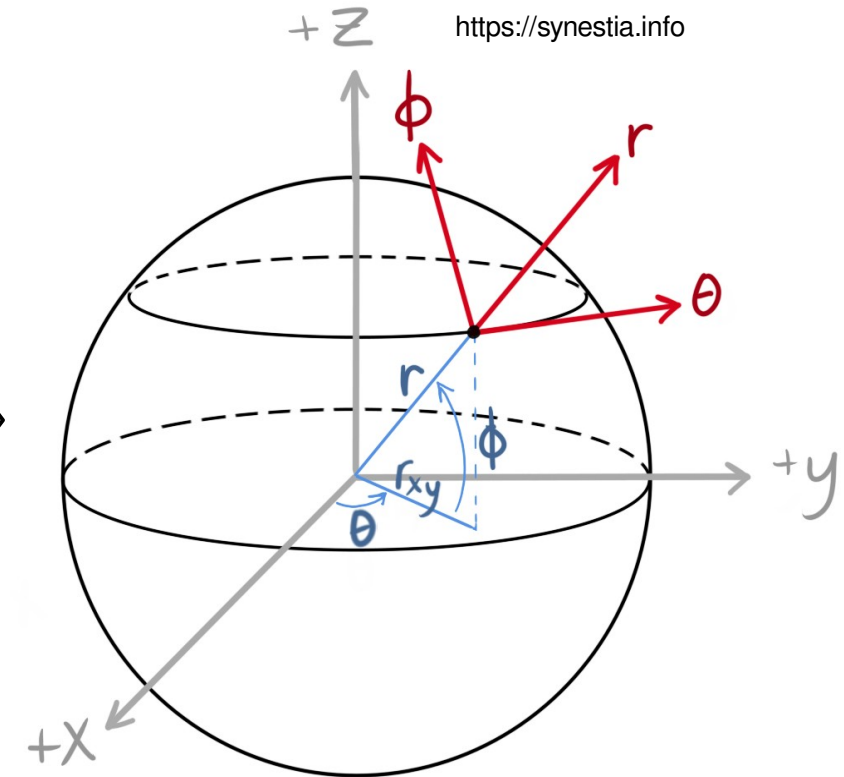
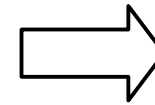
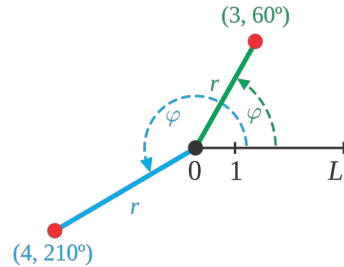
- Structure is defined by a grid, data is a lattice of values
- The field view of Earth
- Similar to raster graphics, but with georeferencing



Coordinate Reference Systems

Geographic coordinates

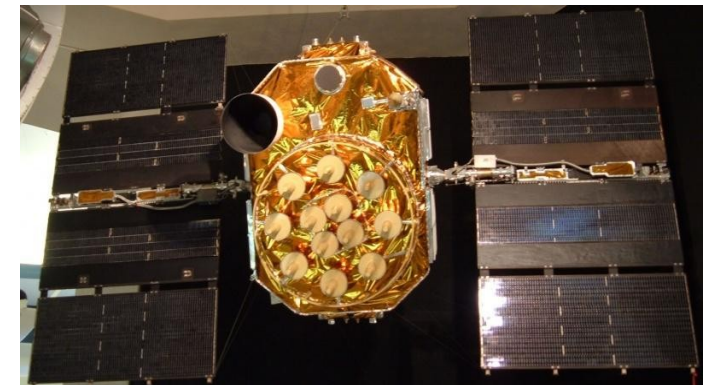
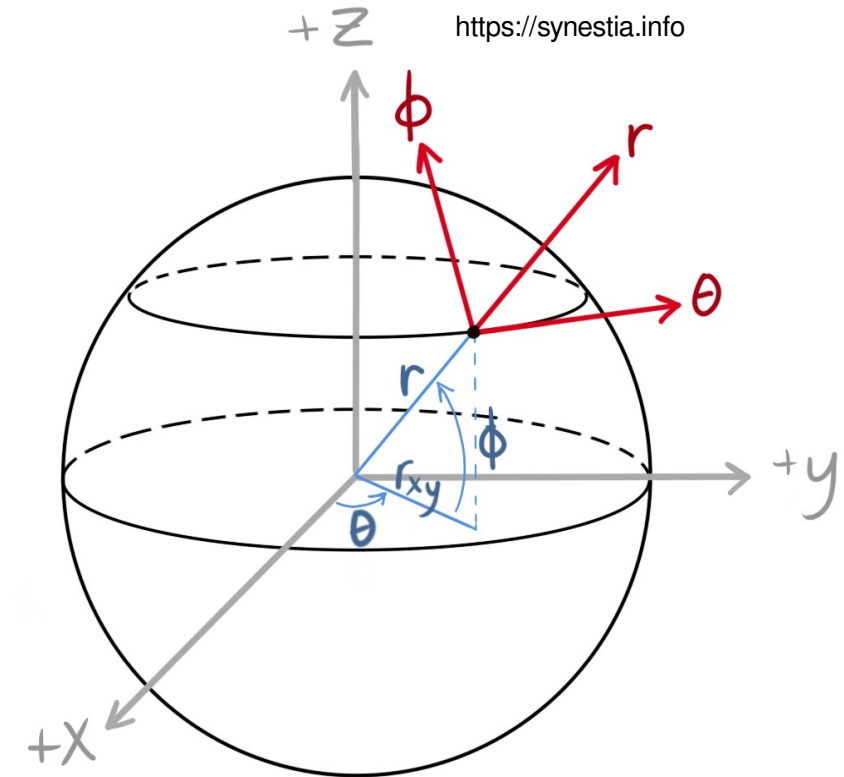
- Based on polar coordinates
- ϕ : latitude
 - $[-90^\circ, 90^\circ]$, equator : 0°
- θ : longitude
 - $[-180^\circ, 180^\circ]$, 0° : prime meridian
- r : elevation



Coordinate Reference Systems

Geocentric coordinates

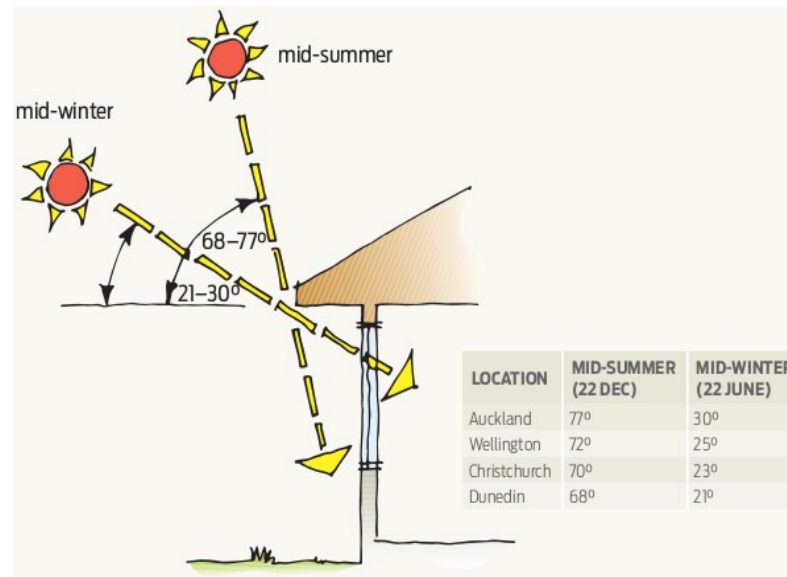
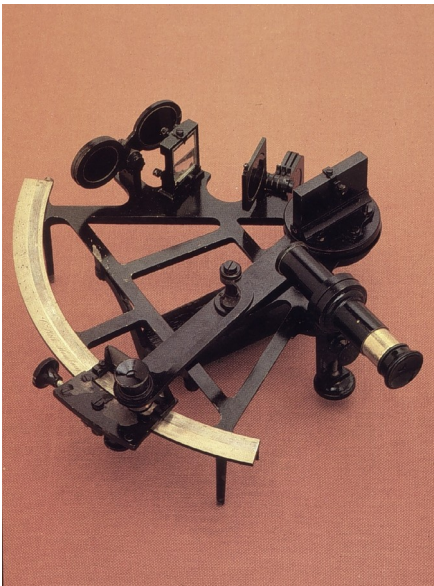
- x, y, z , Earth-Centered Coordinate System
- Based on center of mass, equator and PM
- Used for satellite tracking and navigation (e.g. GPS)
- WGS84 datum



Coordinate Reference Systems

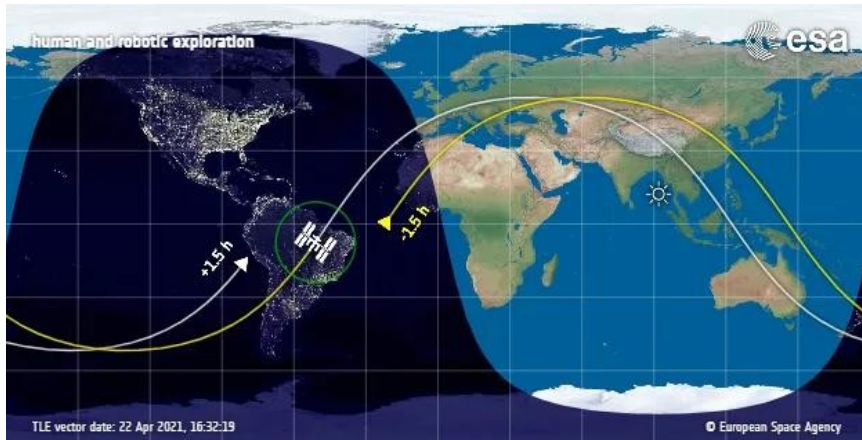
Projection based coordinate systems

- Surface: a 2-dimensional map
- Traditionally: Cartography
- Especially nautical applications

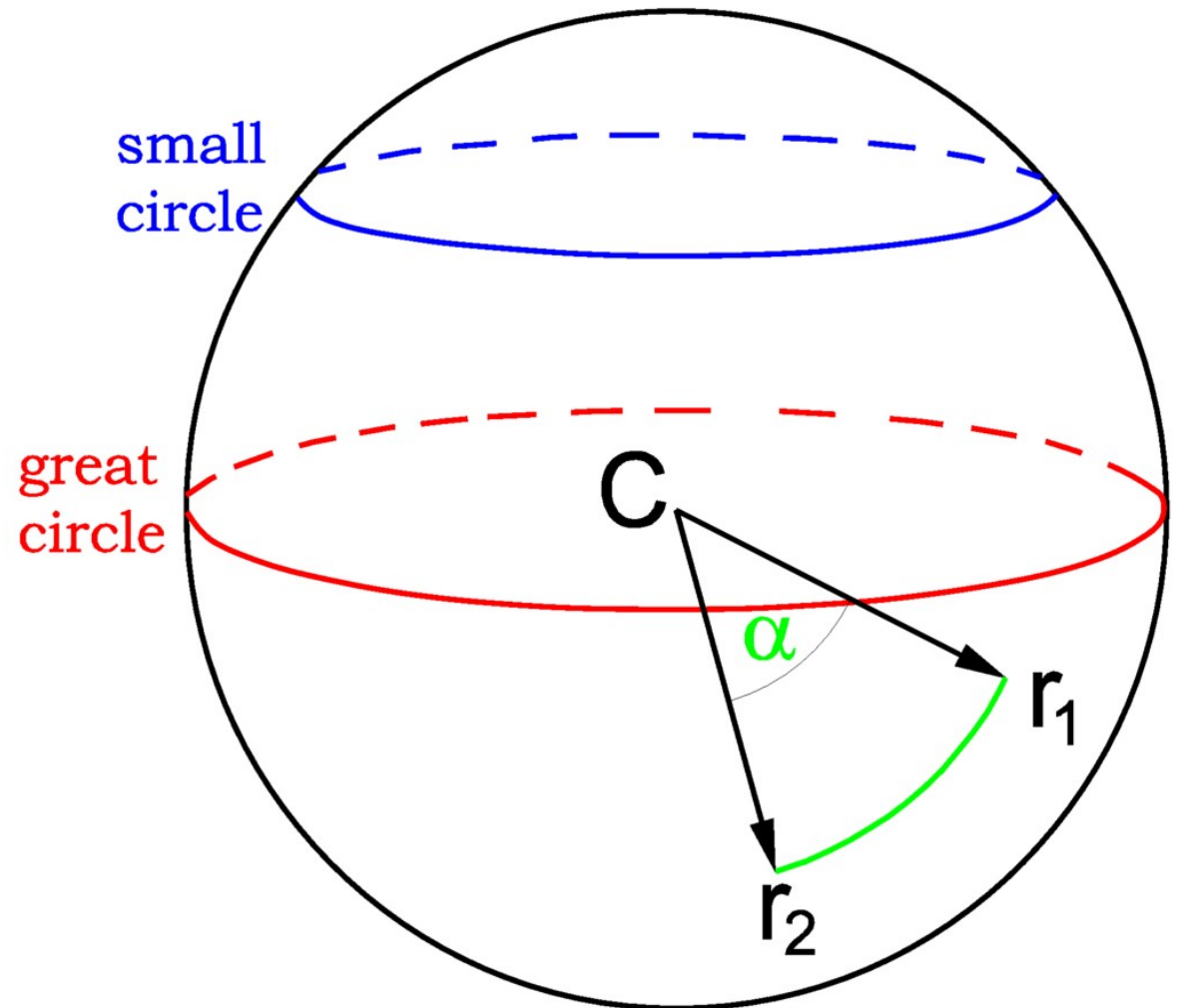


Spheres and Circles

- There are no straight lines
- All points are connected by arcs, either along a great or small circle

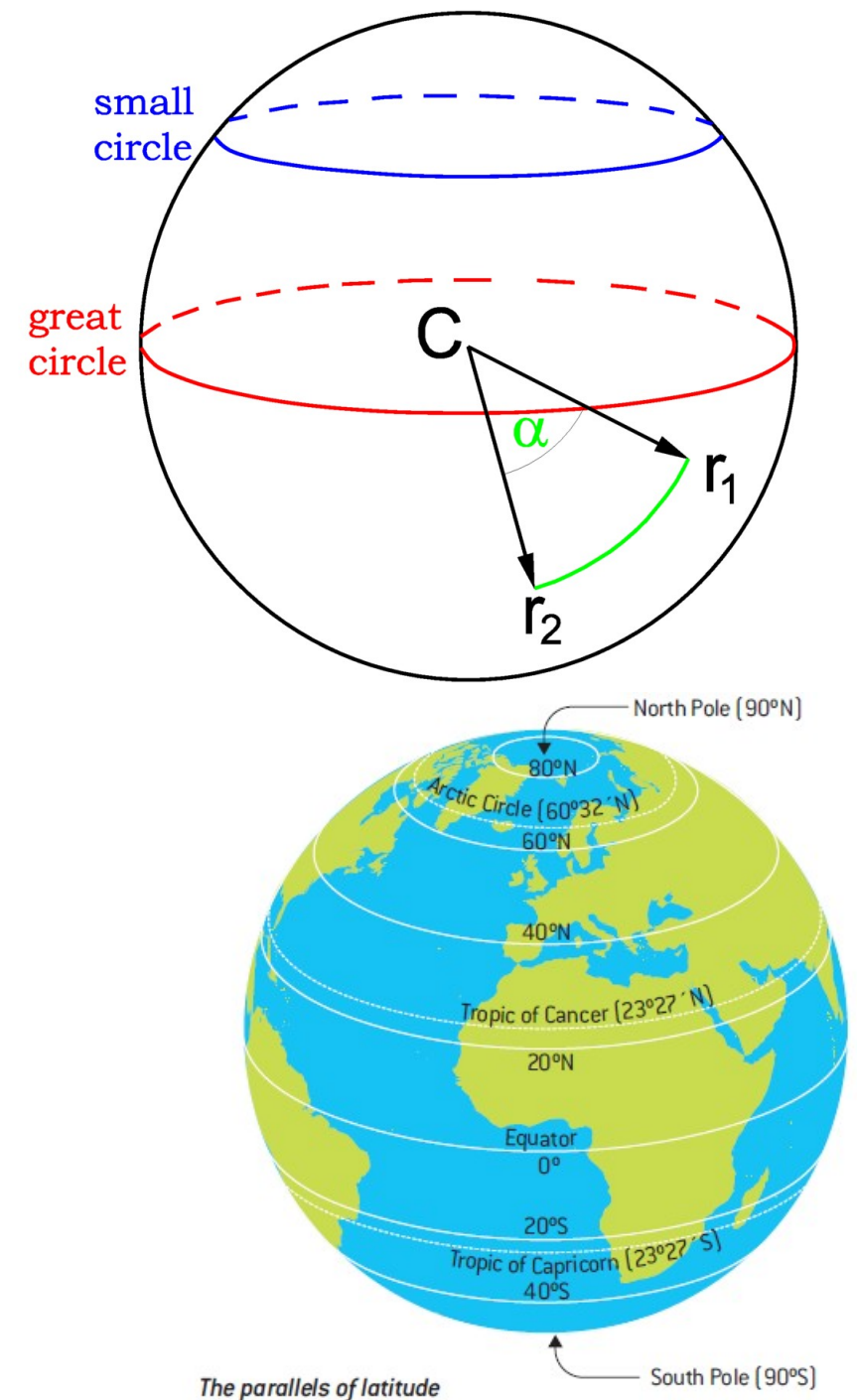


Path of the ISS



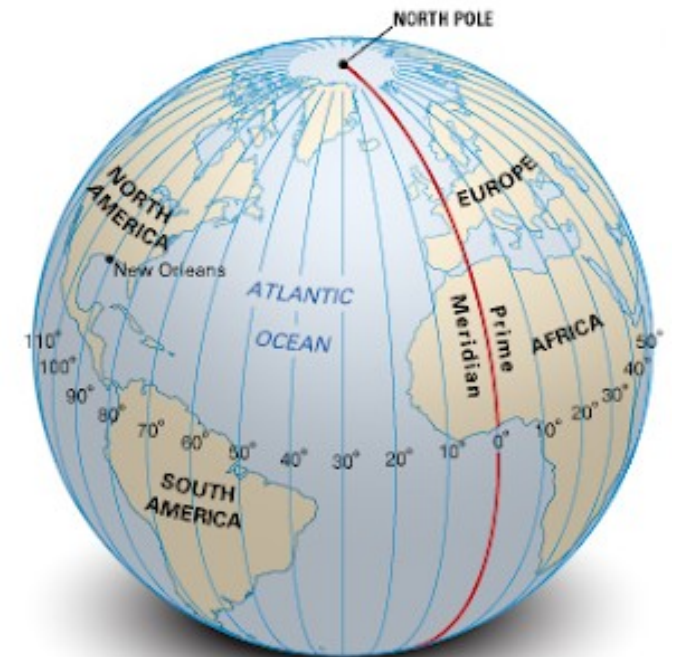
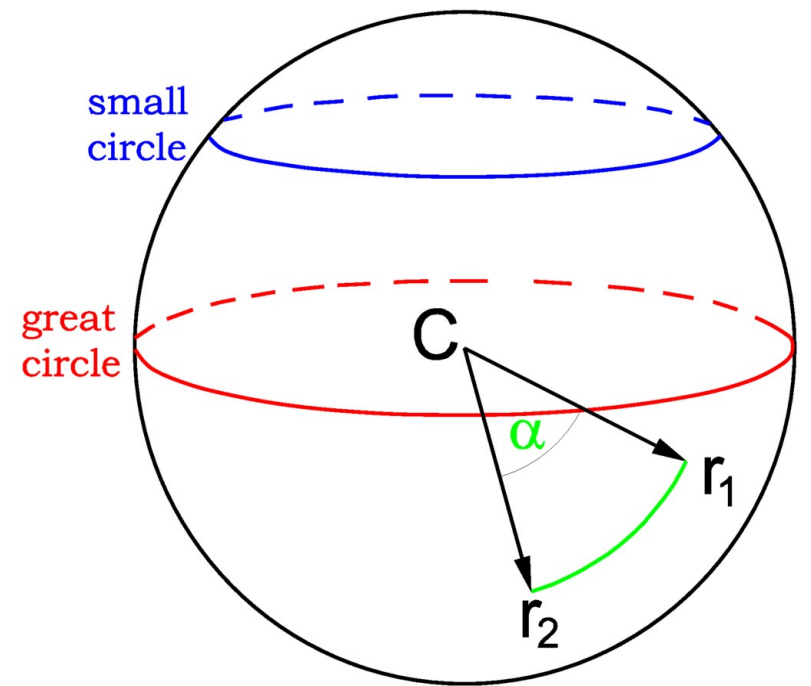
Latitude $[-90, 90^\circ]$

- Latitudinal circles: **parallels**
- The only great circle is the equator!
- Length decrease with a cosine function of latitude
- 10° wide **zone** has different area at different latitudes
- Direction: **zonal (u)**
- Notable circles:



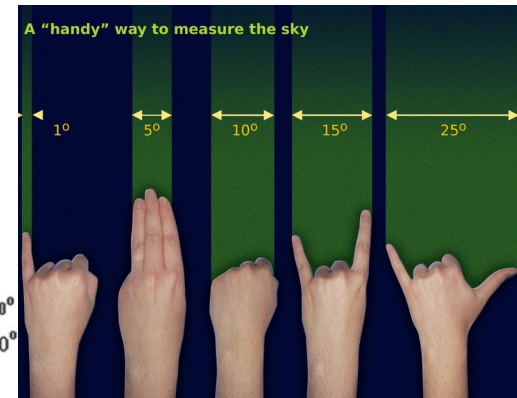
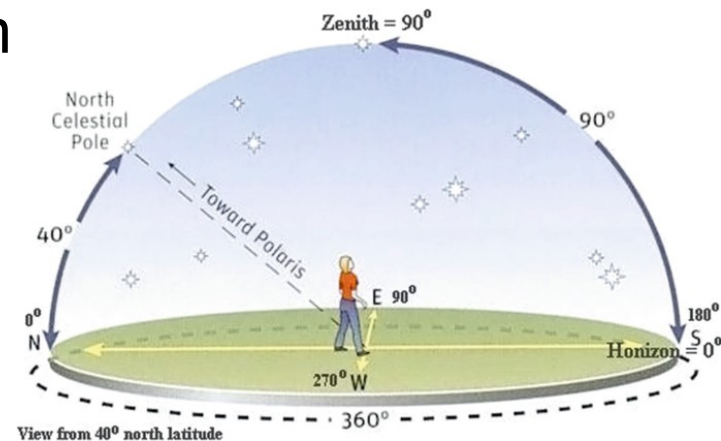
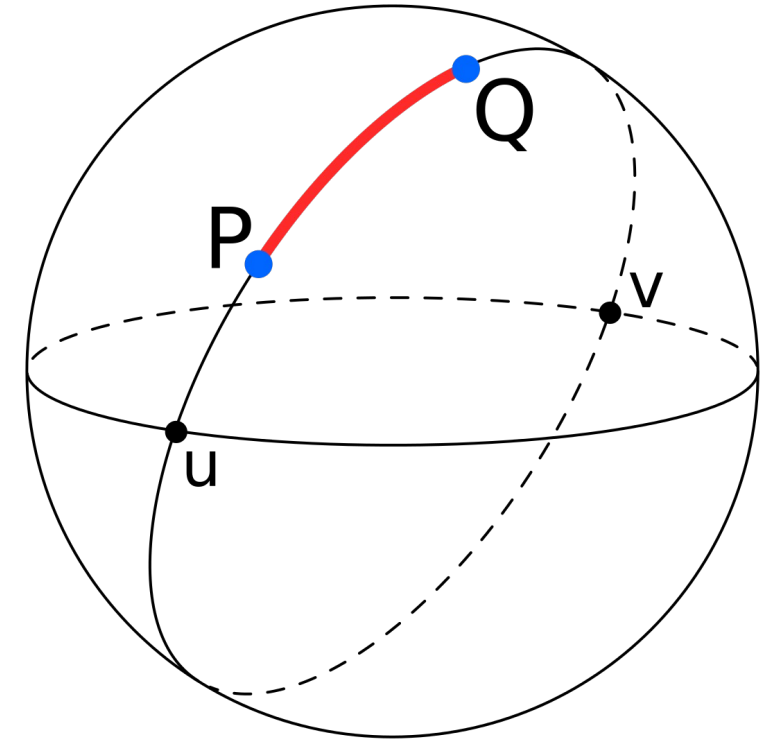
Longitude $[-180, 180^\circ]$

- Latitudinal circles: meridians
- All of them are great circles!
- Their distance decreases at the poles
- Direction: **meridional (v)**
- Notable circle:



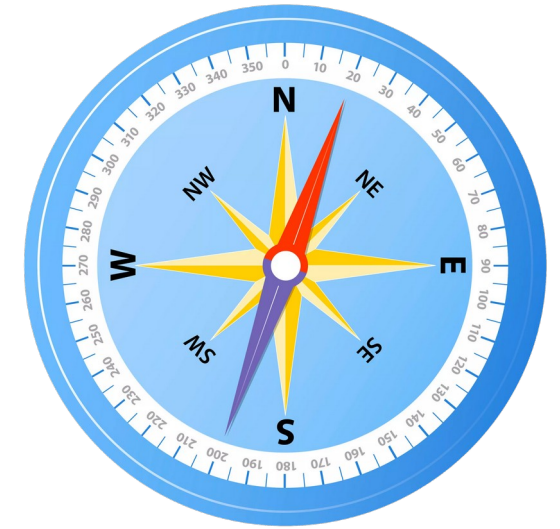
Distances

- Two points define a great circle
- Shortest distance between them is a great circle arc (Great Circle Distance)
- Distance on a surface of a sphere is simplest **in angles (degrees)**
- For practicality we give this in distance:
- 1° (longitude) ~ 111.1 km
- $1'$ (longitude) = 1 nautical mile ~ 1.8 km



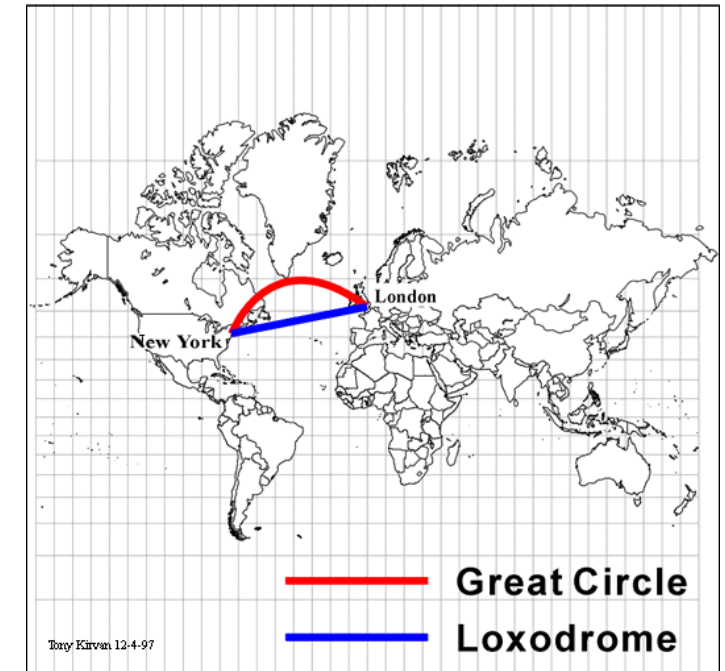
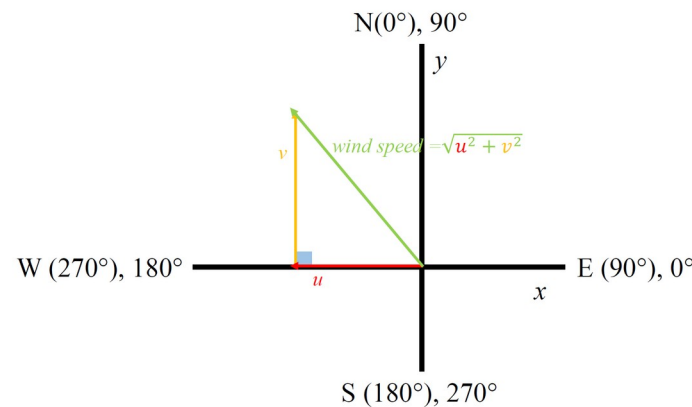
Directions

- Heading: angular distance to north (azimuth)
- Going straight: great circle
- Constant change of heading! (even without the true vs. magnetic difference)



Vectors

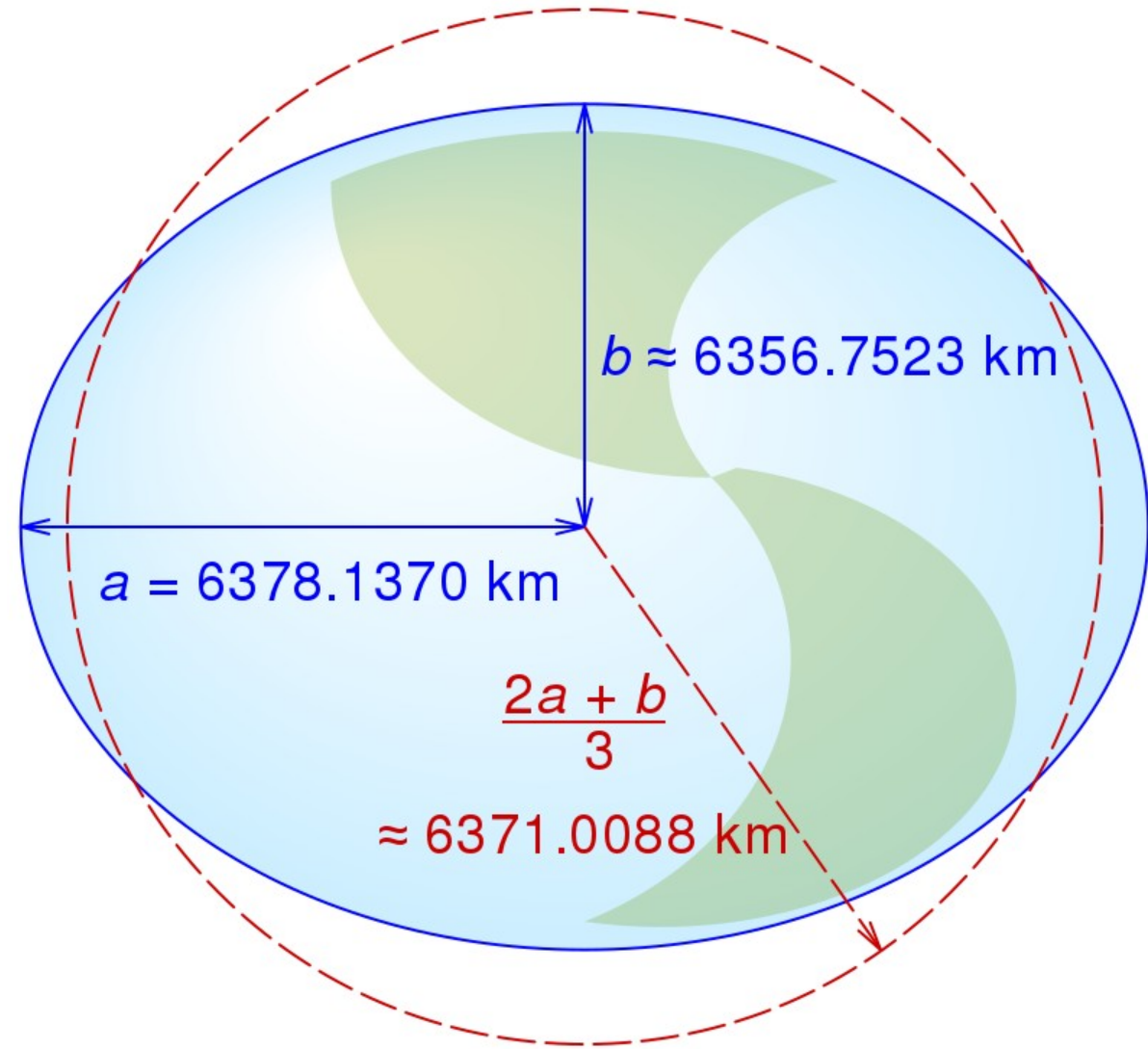
- Given with a) initial heading and magnitude or b) as zonal and meridional components



Mercator projection

The reference ellipsoid

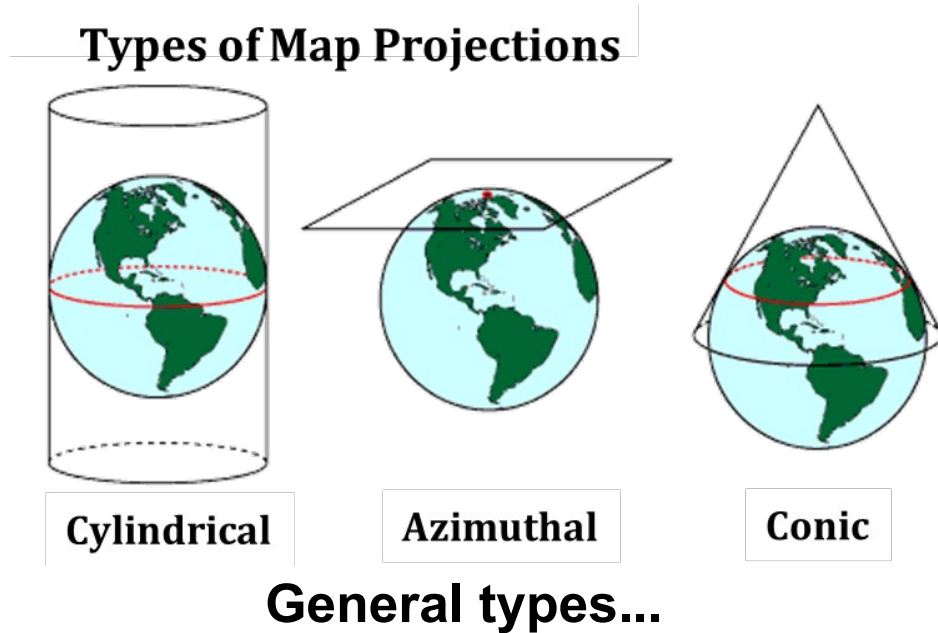
- Approximates the geoid
- Flattened rotational ellipsoid
- Used as a basis for elevation



Coordinate Reference Systems

Map projections

- There is dozens of them
- Complex transformations of spherical coordinates
- They all distort the globe in some sense:
 - conserve area and distort angles
(*equal-area projection*)
 - distort area and conserve angles
(*conformal projection*)
 - distort both area and angles

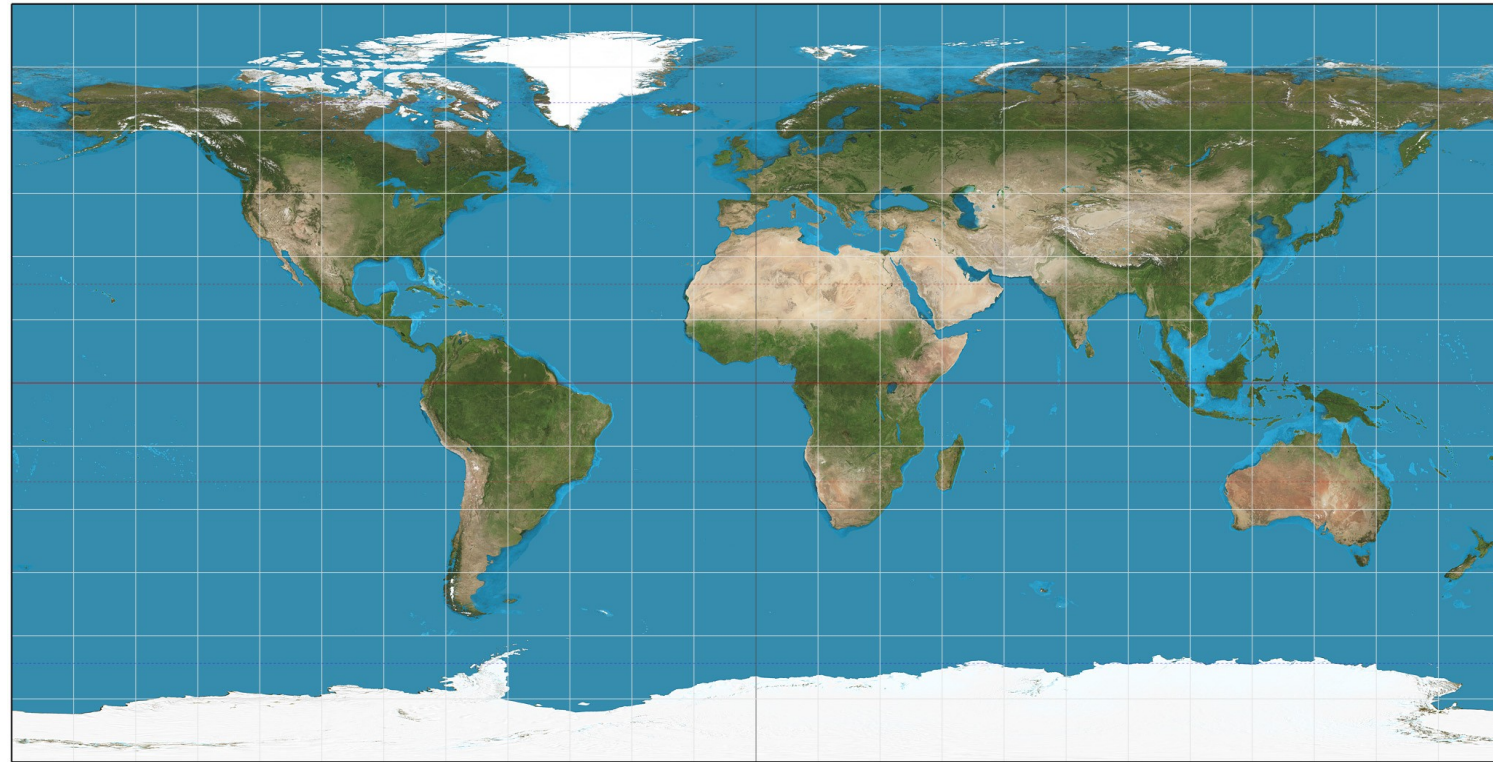


The True size of.. :
<https://www.thetruesize.com>

Coordinate Reference Systems

e.g. Equirectangular Projection

- Longitude (x) – Latitude (y)
- Centered around (0, 0)
- Plate Carée
- Most frequently used for visualization of global-scale data
- Distorts everything



Coordinate Reference Systems

e.g. Mercator projection

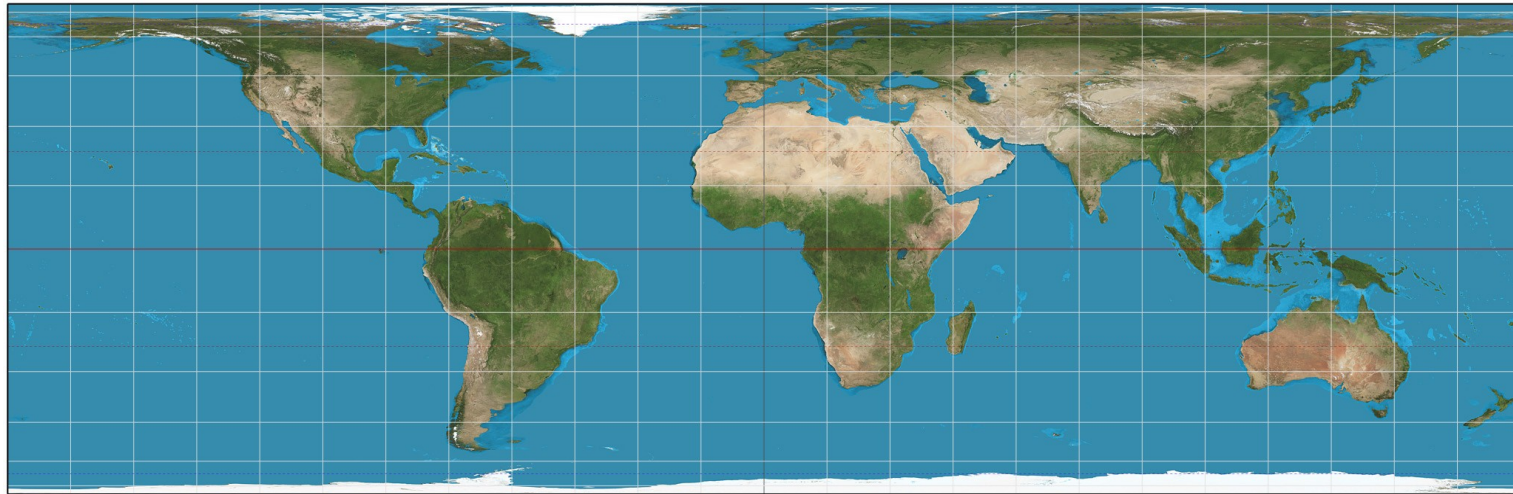
- Conformal
- (nautical use)
- Highly distorts areas
- Cannot represent poles
- Variant: Google Earth



Coordinate Reference Systems

e.g. Lambert's Cylindrical EA Projection

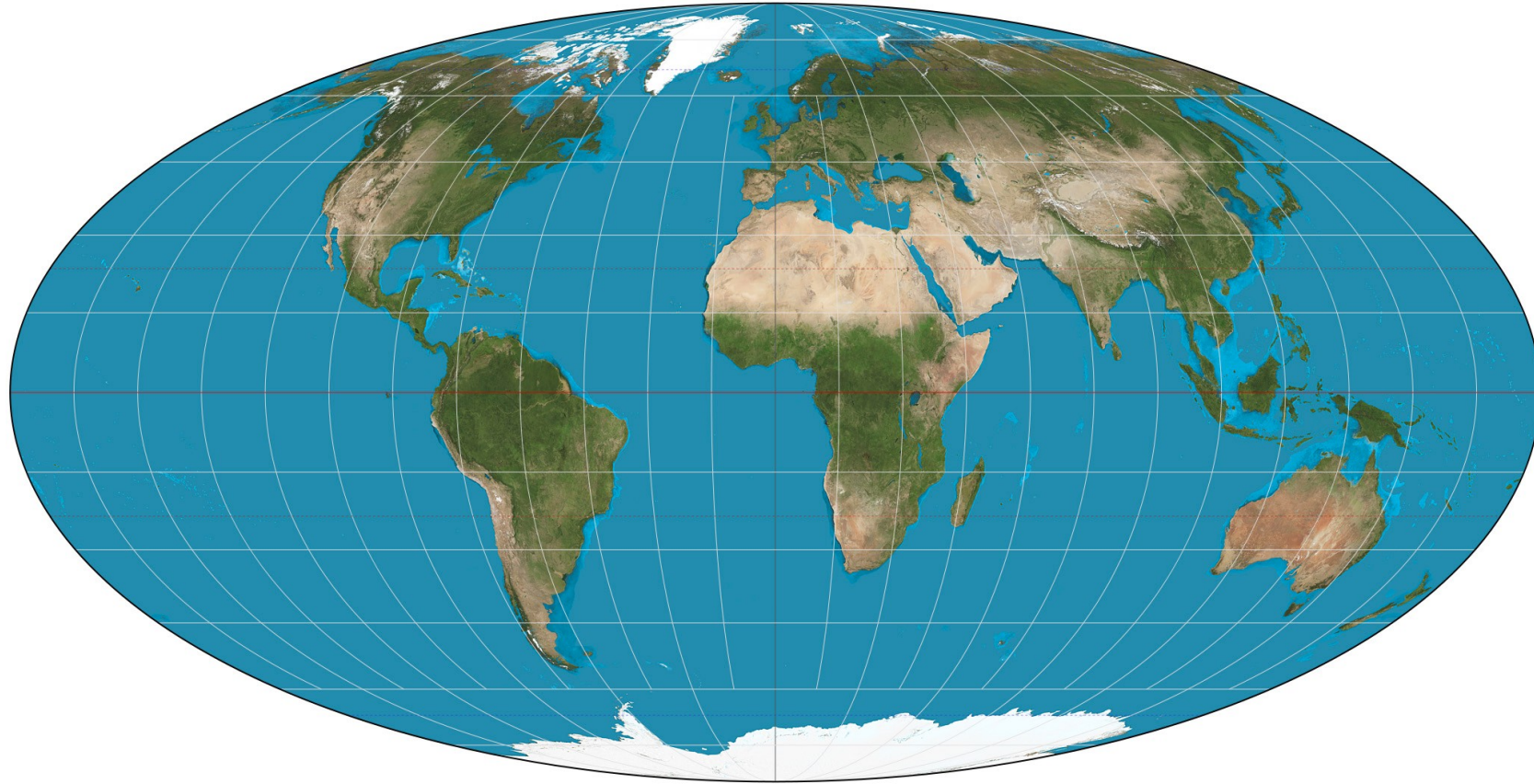
- Equal-area
- Highly distorts angles latitudinally



Coordinate Reference Systems

e.g. Mollweide Projection

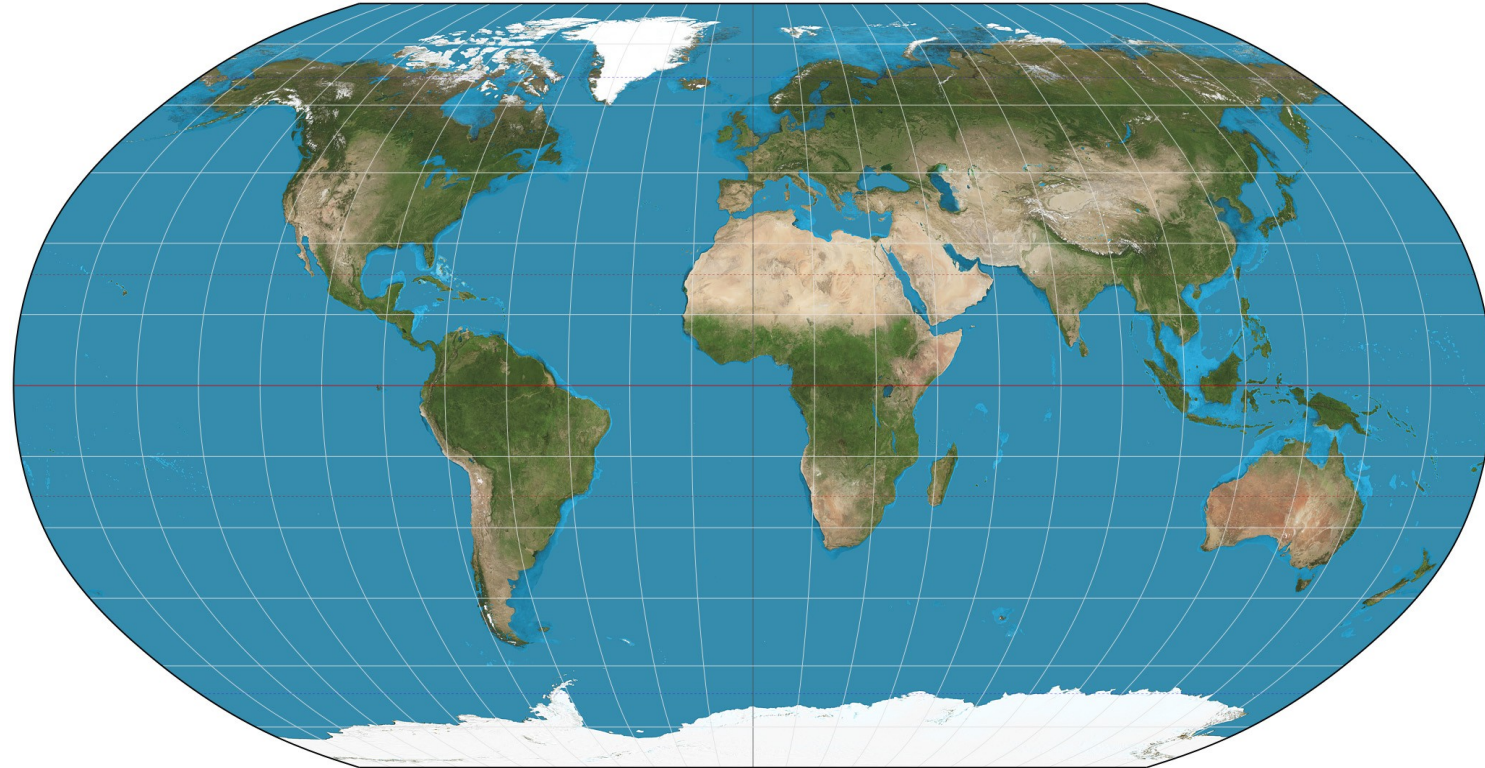
- Equal-area
- Highly distorts angles longitudinally



Coordinate Reference Systems

e.g. Robinson Projection

- Distorts both angles and areas
- Relatively representative
- Promoted by NatGeo and now used by IPCC



Coordinate Reference Systems

Representation

- Older: PROJ.4 string: “+proj=moll”
- Well known text WKT2 string

```
PROJCS["World_Mollweide",  
  GEOGCS["WGS 84",  
    DATUM["WGS_1984",  
      SPHEROID["WGS 84",6378137,298.257223563]],  
    PRIMEM["Greenwich",0],  
    UNIT["Degree",0.0174532925199433]],  
  PROJECTION["Mollweide"],  
  PARAMETER["central_meridian",0],  
  PARAMETER["false_easting",0],  
  PARAMETER["false_northing",0],  
  UNIT["metre",1],  
  AXIS["Easting",EAST],  
  AXIS["Northing",NORTH],  
  AUTHORITY["ESRI","54009"]]
```

- EPSG registry ID (epsg.io) : ESRI:54009



EPSG

GEODETIC PARAMETER DATASET

Managed by IOGP's Geomatics Committee

Implementation

Where, what, why?

GIS tools: Why R?

- Automated manipulation and extraction of data
- Larger quantity of work then with a GUI
- Python is much more powerful
- Integration with other analyses
- Statistical analyses



The Spatial ecosystem



- Multiple R extension packages form the backbone
- The heart: Open Source Geospatial Foundation
- Recipe:
 - **Open source libraries for calculations + R interface**
- R packages hook on the APIs of these libraries
- + extra calculations in compiled code (C, C++, Fortran)

Open Source Libraries/Projects

GDAL: Geodetic Data Abstraction Library

- Definitions of data formats
- Translation between data formats
- <https://gdal.org/>



Open Source Libraries/Projects

PROJ

- Coordinate transformation
- Definition of Coordinate Reference Systems (CRS) and translation
- Current version is 9.2.1 (June 2023)
- R packages rely on older versions (PROJ4 or PROJ6)



Open Source Libraries/Projects

GEOS: Geometry Open Source

- Computational geometry especially towards the manipulation of spatial data
- <http://libgeos.org/>



Open Source Libraries/Projects

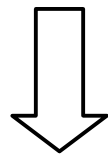
S2 Geometry

- Spherical computations (Google)
- <http://s2geometry.io/>

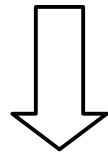


The R packages (two generations)

- Only some are essential, they are on the CRAN
- They are using the libraries
- Class definitions and basic methods
- Lots of packages build on these...



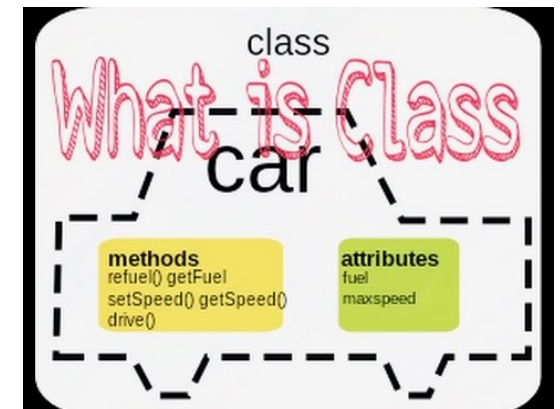
Lot of dependencies, slow development



Generation shift



What is a class?



<https://www.youtube.com/watch?v=XqrkcO42DI8>

Packages

old

new

vector

sp,
rgdal, rgeos

sf, terra

raster

raster

terra, stars

+ others: e.g. geosphere

Installing all of them

Focus on the new but...

- On Windows this is trivial, you can get all from the CRAN with `install.packages()`
- On Mac (and Linux), it is recommended to install them one at a time. They will a) either tell you what libraries need to be installed so they can function properly, 2) only indicate errors and then we have to google :)
- The most important ones:

```
install.packages(c("terra", "sf", "geosphere"))
```

